Unit 2 — Workflow Design, Control Flow & Data Handling in UiPath

(Poornima University, B.Tech AIDS) — Exam Ready Notes

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1. Unit Ke Learning Objectives (Overview)

Isme hum seekhte hain:

- RPA mein **workflow structuring** kaise karte hain.
- **Control flow activities** — If, Switch, Loops, Flowcharts, State Machines.
- **Data handling** — variables, arguments, data tables, collections.
- Efficient automation workflows banane ke liye **sequences aur data operations** ka use.

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2. Workflow Structuring RPA Mein Kyu Important Hai?

Yeh ek **short-answer type** question ban sakta hai — 5 points yaad karo:

1. **Clarity & Readability** — automation samajhna aasan ho jata hai.
2. **Reusability** — components ko dusre projects mein bhi reuse kar sakte ho.
3. **Easier to Maintain & Debug** — workflow achhe se structured ho to error dhundhna aasan.
4. **Error Handling** — exception management smooth ho jata hai.
5. **Scalability** — bade enterprise-level processes ke liye workflow easily scale ho jata hai.

Trick to remember: "**C-R-E-E-S**" — Clarity, Reusability, Easier debug, Error handling, Scalability

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3. Control Flow aur Data Handling — Introduction

Control Flow (kya order mein activities chalti hain):

Activity	**Kaam**
If/Else	Conditional branching (agar/warna)
Switch	Multiple decision paths (bahut saare options mein se ek choose karna)
Loops (While, Do While, For Each)	Repeated execution (baar-baar chalana)
Flowcharts & State Machines	Complex process navigation (complicated paths handle karna)

Data Handling (data kaise store/manage karte hain):

Concept	Kaam
Variables	Single value store karna
Arguments	Data ko ek workflow se dusre workflow mein pass karna
DataTables	Structured data handle karna (jaise Excel)
Collections (Lists, Dictionaries)	Dynamic datasets manage karna

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4. Sequences aur Data Operations Ki Relevance

Sequences:

- **Linear workflows** hote hain — step by step top se bottom execute hote hain.
- Simple, straightforward tasks ke liye suitable.
- Bade workflows (Flowchart/State Machine) ke **building blocks** ke roop mein use hote hain.

Data Operations:

- Real-world business automation enable karte hain.
- Examples: **filter, sort, merge, transform, export data**.
- Zaruri hain invoice processing, report generation, email automation jaise tasks ke liye.

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5. Sequence — Detail Mein (Definition + Kab Use Kare)

Definition:

Sequence UiPath ka **sabse simple workflow type** hai. Activities ko **linearly, step-by-step** top se bottom execute karta hai.

Kab Use Kare Sequence:

- Jab process ka **clear, linear flow** ho, bahut decision branches na ho.
- Chhote tasks ke liye (e.g., read data → process data → write results).
- Bade workflows (Flowchart/State Machine) ke andar **building blocks** ke roop mein.

Example Use-Cases:

- App mein login karna
- Excel mein data read/write karna
- Email notification bhejna

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6. Flowchart — Detail Mein (Definition + Best Use-Cases)

Definition:

Flowchart ek workflow type hai jo **non-linear process design** allow karta hai. **Nodes aur connectors** use karke different execution paths define karte hain. Complex processes ke liye suitable hai jinme multiple decision points hote hain.

Best Use-Cases:

- Processes jinme **multiple branches/outcomes** hote hain.
- Jab decisions **conditions ya user inputs** pe depend karte hain.
- **Transactional workflows** ke liye jinme multiple checks chahiye.

Example Use-Cases:

- Customer support ticket resolution (issue type ke basis pe route karna)
- Order processing (New Order, Return, Cancel — alag-alag flow)
- Approval workflows (Manager approval vs Auto-approval)

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7. Sequence vs Flowchart vs State Machine (BAHUT IMPORTANT COMPARISON TABLE)

Feature	Sequence	Flowchart	State Machine
Structure	Linear, step-by-step	Graphical, branch-based	State-driven with transitions
Complexity	Simple	Moderate to High	High (best for long-running)
Best For	Small tasks, building blocks	Complex decision-making	Large, rule-based workflows
Readability	Very easy	Easy to visualize branching	Good for event/state handling
Examples	Login to app, Read/Write file	Order processing, Routing tasks	Order lifecycle, multi-state processes
Execution Flow	Ek hi path chalti hai	Multiple paths, flexible	Defined states ke beech move karta hai

Yaad rakhne ka concept: Sequence = "seedha rasta"; Flowchart = "branch wala rasta"; State Machine = "states ke beech jump karna" (jaise Order ka status: Placed → Shipped → Delivered).

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8. Simple Workflow Design — Sequence + Flowchart Examples (Practical Question)

Example A: Sequence-Based Workflow

Task: Excel se invoice data read karo aur email send karo.

Steps in Sequence:

1. Excel file read karo → DataTable mein store.
2. Har row (invoice) process karo.

3. Message string generate karo.
4. Email send karo.

Flow: Linear (no branching)

Example B: Flowchart-Based Workflow

Task: Customer order processing.

Steps in Flowchart:

1. Input: Order request.
2. **Decision Node:** Order type check karo.
    - Agar **New Order** → Payment process karo → Product ship karo.
    - Agar **Return** → Validate karo → Refund process karo.
    - Agar **Cancel** → Shipment stop karo → Agar paid hai to refund karo.

Flow: Branching and conditional paths

Summary:

- Sequences → Linear, simple workflows
- Flowcharts → Branching, decision-heavy workflows
- State Machines → Large, rule-based, long-running processes
- Teeno milke UiPath workflow design ke **core building blocks** hote hain.

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9. Control Flow Concepts — Detail Mein

Definition:

Control flow define karta hai ki workflow mein activities **kis order mein execute** hongi.

Control flow bots ko allow karta hai:

- **Decide** karna — kaunsa path lena hai (conditions)
- **Repeat** karna — activities repeat karna (loops)
- **Branch** karna — multiple directions mein (Switch, Flowchart)
- **Parallel** tasks execute karna (jaha possible ho)

Important line: Control flow ke bina, workflows sirf **linearly** execute hote, bina koi "intelligence" ke.

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10. Conditional Branching — If Activity

If Activity:

- Agar condition **true** hai to ek set of activities execute hoti hai, agar **false** hai to dusra set.

- Syntax: If (Condition) → Then / Else

Example:

If (invoiceAmount > 10000)

→ Send to Manager Approval

Else

→ Auto Approve

Simple samjho: "Agar X ho to Y karo, warna Z karo" — bas yahi If activity hai.

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11. Switch Activity

- Jab **multiple possible outcomes** hote hain, tab use karte hain.
- Multiple nested Ifs se **cleaner** hota hai (code neat rehta hai).

Example:

Switch (DocumentType):

Case "Invoice" → Process Invoice

Case "Receipt" → Process Receipt

Case "PO" → Process Purchase Order

Simple samjho: If activity 2 options (true/false) ke liye use hoti hai, jab options 3+ ho to Switch use karo.

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12. Loops (Bahut Important — 4 Types Yaad Karo)

While Activity

- Jab tak condition **true** hai, activities repeat hoti rahengi.
- Condition **execution se pehle** check hota hai (pre-check).
- Example: While (counter < 5) → Perform Data Entry.

Do While Activity

- While jaisa hi hai, lekin condition **execution ke baad** check hoti hai (post-check).
- Isliye block **kam se kam ek baar** zarur chalega.
- Example: Do → Try Login → While (LoginSuccess = False).

For Each Activity

- Collections (List, Array, DataTable) pe **iterate** karta hai.
- Example: For Each email in emailList → Send Email.

Parallel Activity

- Multiple activities **ek saath simultaneously** execute karta hai.
- **Independent tasks** ke liye best hai.
- Example: Parallel branches → Download Report + Send Reminder Email

Comparison Table (Loop Types):

Loop Type	Kab Use Kare	Example Use-Case
While	Condition fail hone tak repeat (pre-check)	Login attempts retry karna success tak
Do While	Kam se kam ek baar execution ensure karna (post-check)	Ek baar "Refresh" click karke, phir data check karna
For Each	Collection pe iterate karna	Excel ki saari rows ya inbox ke saare emails loop karna
Parallel	Independent tasks ek saath chalana	2 apps se ek saath data extract karna

Trick to remember: "**W-D-F-P**" — While (pre-check), Do while (post-check, at least once), For each (collection), Parallel (simultaneous)

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13. Decision Making — Real-World Automation Examples

Scenario	Logic Used
Invoice Processing	If invoice > ₹10,000 → Manager approval, else auto-approve
Email Classification	Switch(EmailSubject): "Payment"→Finance folder, "Job Application"→HR folder
Data Scraping	While → jab tak "Next Page" button disabled na ho tab tak scrape karo
System Login	Do While → login attempt karo jab tak LoginSuccess = True na ho
Bulk Notifications	For Each email → personalized notification send karo

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14. Step-by-Step Combined Examples (Sequence + Control Flow) — Important Practical Q

Example A: Processing Student Results

Sequence + If + For Each

1. Excel file read karo (students + marks).
2. For Each student row →
    - If (Marks ≥ 40) → "Pass" assign karo, warna "Fail".
3. Results ko Excel mein wapas write karo.

Example B: Web Data Extraction

Sequence + While

1. Website open karo.
2. While (NextPage button exist karta hai):
    - Table data extract karo.
    - Next Page click karo.
3. Sare results ko ek DataTable mein merge karo.

Example C: Order Handling

Sequence + Switch

1. Order type get karo.
2. Switch (OrderType):
    - Case "New" → Order create karo.
    - Case "Cancel" → Shipment stop karo.
    - Case "Return" → Refund start karo.

Summary:

- Control flow automation mein **intelligence** add karta hai.
- **If & Switch** → Decision making
- **While, Do While, For Each** → Looping
- **Parallel** → Simultaneous execution
- Real-world automation mein hamesha **Sequences + Control Flow** combine hote hain flexibility aur efficiency ke liye.

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15. Variables — Detail Mein (VERY IMPORTANT SECTION)

Definition:

Variable ek **named memory location** hai jo automation ke time data store karne ke liye use hoti hai. Variables automation ko **dynamic** banate hain — data reuse, update, ya activities ke beech pass ho sakta hai. Variables ke bina automation **static** (hard-coded values) rehta.

Types of Variables (8 common types — yaad karo):

Type	Kya Store Karta Hai	Example
String	Textual data	"Hello, UiPath!"
Int32	Integer numbers (whole numbers)	count = 25

Type	Kya Store Karta Hai	Example
Boolean	True/False values	isLoggedIn = True
Double	Decimal numbers	price = 1250.75
DateTime	Date aur time values	invoiceDate = 12-Sep-2025
Array/List	Items ka collection	string[] emails = {"a@gmail.com", "b@gmail.com"}
DataTable	Structured, tabular data	Excel data import karna
Dictionary<TKey,TValue>	Key-value pairs	{"EmpID: 101", "Name: Ravi"}

Trick to remember: "**S-I-B-D-D-A-D-D**" (String, Int32, Boolean, Double, DateTime, Array, DataTable, Dictionary)

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16. Variable Naming Conventions & Data Types

Naming Conventions:

- **Meaningful Names** do — Bad: var1, temp | Good: customerName, invoiceAmount
- **CamelCase/PascalCase** use karo — Example: userEmail, TotalAmount

Choosing Data Types:

- Hamesha **specific data type** use karo, GenericValue ke bajaye.
- Isse improve hota hai:
    - **Performance** (execution fast hota hai)
    - **Readability** (developers easily samajh jate hain)
    - **Debugging** (type errors early catch ho jate hain)

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17. Variable Scope — Sequence, Flowchart, Global (Important Concept)

Scope define karta hai ki variable workflow mein **kaha access** ho sakta hai.

Types of Scope:

1. **Sequence-Level Scope**
    - Variable sirf **usi sequence ke andar** available hota hai.
    - Example: rowCounter sirf "ProcessInvoice.xaml" mein use hoga.
2. **Flowchart-Level Scope**
    - Variable **flowchart ke sabhi branches** mein accessible hota hai.
    - Example: orderStatus different decision nodes mein accessible.

3. **Global Scope**
    - Variables **Main.xaml (highest level)** pe define hote hain.
    - Pure project mein accessible hota hai.
    - **Cross-workflow data transfer** ke liye Arguments use karna better hai, bahut saare global variables ke bajaye.

Yaad karne ka trick: Scope jitna "chhota" hoga, workflow utna clean aur error-free rahega.

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18. Variables Efficiently Use Karne Ke Best Practices (Exam mein "best practices likho" wala Q)

1. **Arguments Use Karo Workflow Communication Ke Liye**
    - Global variables ke bajaye, **In, Out, In/Out arguments** use karo.
    - Example: invoiceID ko ReadInvoice.xaml se ProcessInvoice.xaml mein pass karo.
2. **Scope Limit Karo**
    - Variables ko wahi scope do jaha unki zarurat hai.
    - "Project-wide" variables avoid karo jab tak zaruri na ho.
3. **Properly Initialize Karo**
    - Default values set karo runtime errors avoid karne ke liye.
    - Example: intCounter = 0, strMessage = ""
4. **GenericValue Overuse Avoid Karo**
    - Generic flexible hai lekin type conversion mein errors de sakta hai.
    - Example: "123" (string) vs 123 (integer) confusion.
5. **Clarity Ke Liye Naming**
    - Consistent prefixes use karo (str, int, dt, dict) collaboration easy karne ke liye.
6. **Variables Reuse Karo**
    - Unnecessary naye variables mat banao.
    - Example: Agar dtInvoices already hai, to filter karne ke liye usi ko reuse karo, dtInvoicesFiltered banane ke bajaye (jab tak zaruri na ho).

Summary:

- Variables dynamic automation ke liye **essential** hain.
- UiPath multiple data types support karta hai (String, Int32, Boolean, DataTable, etc.)
- Clear naming conventions aur right data type reliability ke liye use karo.
- Scope control karo (sequence, flowchart, global) errors avoid karne ke liye.

- Arguments use karo workflows ke beech data pass karne ke liye, GenericValue overuse mat karo.

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19. Arguments — Detail Mein (VERY IMPORTANT — Variables se differentiate karna aa sakta hai)

Definition:

Arguments **data carriers** hain jo UiPath mein ek workflow se dusre workflow mein information pass karne dete hain. Variables workflow ke **andar** data store karte hain, arguments workflows ke **beech (in/out)** data transfer enable karte hain. **Modular, reusable workflows** banane ke liye useful hain.

Types of Arguments (Direction ke basis pe):

1. **In**
    - Data ko workflow mein **pass** karta hai.
    - Workflow value **read** kar sakta hai, **modify nahi** kar sakta.
    - Example: invoiceID ko ProcessInvoice.xaml mein pass karna.
2. **Out**
    - Data ko workflow se **bahar** pass karta hai.
    - Example: Extracted totalAmount calling workflow ko wapas bhej diya jata hai.
3. **In/Out**
    - Reading aur writing dono allow karta hai.
    - Example: statusFlag workflow mein pass hota hai, andar update hota hai, aur wapas return hota hai.

Trick to remember: "In = andar jaye, Out = bahar aaye, In/Out = dono"

Reusability of Workflows with Arguments:

- Arguments workflows ko **modular aur reusable** banate hain.
- Instead of same steps multiple jagah likhne ke:
    1. Ek workflow banao (e.g., Login.xaml)
    2. Arguments add karo (username, password)
    3. Different values pass karke different automations mein reuse karo

Example:

- SendEmail.xaml arguments ke saath: In: recipientEmail, In: subject, In: body
- HR, Finance, aur Support processes mein different values ke saath reuse ho sakta hai

Variable vs Argument — Quick Difference:

Point	Variable	Argument
Scope	Ek hi workflow ke andar	Workflows ke beech (cross-workflow)
Use	Internal data store karna	Data pass karna (communication)

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20. Collection Types — Arrays, Lists, Dictionaries (Important comparison)

Collections **multiple values** store karte hain aur automation mein widely use hote hain.

Array

- **Fixed-size** collection, same data type ke items ka.
- Example: string[] emails = {"a@gmail.com", "b@gmail.com"}
- **Static data** ke liye acha hai jaha items ki number pehle se pata ho.

List<T>

- **Dynamic collection** (grow ya shrink ho sakta hai).
- Example: List of Int32 {10, 20, 30}
- Arrays se zyada **flexible** hota hai.
- Loops aur filtering operations mein common use hota hai.

Dictionary<TKey, TValue>

- **Key-value pairs** store karta hai.
- Example: Dictionary(Int32, String) = {101 → "Ravi", 102 → "Meena"}
- **Lookup tables** ke liye useful (jaise EmployeeID → EmployeeName)

Trick to remember: Array = "fixed box", List = "flexible bag", Dictionary = "label wala box (key-value)"

For Each with Collections — Practical Use Cases:

1. **For Each (Array of Emails)** → Har address pe email send karo
2. **For Each (List of File Paths)** → Har file read/process karo
3. **For Each (Dictionary of Employees)** → Key=EmployeeID, Value=Name → payroll data display/process karo
4. **For Each Row in DataTable** → Structured Excel data process karo (invoices, marks, orders)

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21. Clipboard Activities

Definition:

Clipboard activities automation ko **system clipboard** (copy-paste memory) ke saath interact karne dete hain.

Activities:

- **Get From Clipboard:** Clipboard mein currently stored text **retrieve** karta hai.
    - Example: Web form ya Excel cell se text copy karke automation mein use karna.
- **Set To Clipboard:** Given text ko clipboard mein **place** karta hai.
    - Example: Invoice number ko clipboard mein copy karna ERP system mein paste karne ke liye.

Usage Scenarios:

- Apps ke beech copy-paste operations automate karna
- Ek application se values extract karke dusre mein paste karna
- Bina file save kiye quick text transfer

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22. File Operations — Read, Write, Append (Important)

Operation	Kaam	Example
Read Text File	.txt file ka content ek **string variable** mein read karta hai	Configuration details ya template emails load karna
Write Text File	Text ko naye ya existing file mein likhta hai (**overwrite** karta hai existing content ko)	Scraped data ya generated reports save karna
Append Line	Existing file ke **end mein** text add karta hai, bina overwrite kiye	Transactions ke logs rakhna, multiple entries store karna

Simple samjho: Write = "purana content mita ke naya likhna"; Append = "purane content ke aage naya jodna"

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23. File Handling — Move, Copy, Delete

- **Move File:** File ko ek location se dusri location mein **move** karta hai.
    - Example: Processed invoices ko "Input" folder se "Archive" mein move karna.
- **Copy File:** File ki ek **duplicate** naye location pe banata hai.
    - Example: Processing se pehle reports ka backup lena.

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24. Real-World File-Process Automation Scenarios (Exam mein application-based Q aa sakta hai)

Scenario 1 — Log Automation:

- **Append Line** use karo har automation step record karne ke liye (success, errors, timestamps).
- Isse debugging aur audit trails mein help milti hai.

Scenario 2 — Report Generation:

- Excel ya database se data read karo.
- Results ko .txt ya .csv report files mein write/append karo.
- Reports ko shared folder mein move karo ya email attachment banao.

Scenario 3 — File Management:

- Ek hafte baad automatically purane log files clean karo.
- Daily processed files backup drive mein copy karo.
- Successfully processed documents "Archive" mein move karo compliance ke liye.

Summary:

- Clipboard activities → copy-paste automation enable karti hain
- File operations → text files ka read/write/append enable karte hain
- File handling → files ko systematically move, copy, delete karta hai
- Real-world use → logs, reporting, invoice/document automation ke liye widely use hota hai

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25. Excel/CSV ↔ DataTable Operations (BAHUT IMPORTANT — practical exam question)

Excel Activities:

- **Read Range:** Excel sheet ka data ek **DataTable** mein read karta hai.
- **Write Range:** DataTable ko Excel mein write karta hai.
- **Read Cell / Write Cell:** Single cell read/write karta hai.

CSV Activities:

- **Read CSV:** .csv file ko DataTable mein read karta hai.
- **Write CSV:** DataTable ko .csv format mein export karta hai.

Best Practices:

- **Excel** ke liye **Excel Application Scope** use karo jab automation Excel-specific features (macros) pe depend kare.
- **CSV** ke liye proper delimiters (, ya ;) ka use ensure karo.

Converting Excel/CSV ↔ DataTable:

- **To DataTable:** Read Range / Read CSV → directly DataTable output karta hai. Bulk processing (filter, sort, loop) ke liye useful.
- **From DataTable:** Write Range / Write CSV → DataTable ko wapas file mein convert karta hai. Example: 10,000 rows process karke, filter karke, sirf needed records export karna.

DataTable Filtering, Sorting, Looping:

Operation	Activity	Example
Filtering	Filter DataTable	Status = "Approved" wali rows hi rakho
Sorting	Sort DataTable	Employees ko Salary se sort karo (ascending/descending)
Looping	For Each Row in DataTable	row("ColumnName").ToString se access karo, sab student records loop karo

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26. Break, Continue (Pass), Else in Control Structures

- **Break:** Loop se **immediately exit** kar deta hai.
    - Example: First match milte hi searching stop karna.
- **Continue (UiPath mein "Pass" kehte hain):** Current iteration **skip** karke next pe chala jata hai.
    - Example: Missing data wali rows skip karna.
- **Else:** Decision-making (If activity) mein **alternate path** define karta hai.
    - Example: If InvoiceAmount > 5000 → Approve; Else → Reject

Trick to remember: Break = "poori loop band karo"; Continue/Pass = "iss iteration ko skip karo, aage badho"

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27. Hands-On Example: Read Excel → Filter Data → Write CSV (Important Practical Q)

Scenario: Employee salary Excel file process karo aur sirf high-salary employees ko CSV mein export karo.

Steps:

1. **Read Range** (Excel Application Scope) → data ko dtEmployees mein store karo.
2. **Filter DataTable:**
    - Condition: Salary > 50000
    - Output → dtHighSalary
3. **Write CSV** → dtHighSalary ko HighSalaryEmployees.csv mein export karo.

Result:

- Input: Employees.xlsx with 500 rows
- Output: CSV file jisme sirf salary > 50,000 wale employees hain

Summary:

- Excel/CSV ↔ DataTable RPA data automation ka **backbone** hai.
- Filtering, sorting, looping bade datasets process karne mein help karte hain.

- Break/Continue/Else workflows ko **flexible aur efficient** banate hain.
- Real-world use → payroll, invoices, reports, compliance data automation.

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28. FINAL RECAP — Sabse Important Points (Exam Se Pehle Yahi Padho)

Sequence, Flowchart, Control Flow:

- **Sequence** → Linear, step-by-step tasks (simple processes ke liye best)
- **Flowchart** → Visual, non-linear (decision-based workflows ke liye best)
- **Control Flow** → If, Switch, Loops (While, For Each), Parallel (decisions & repetition ke liye)

Variables/Arguments & Data Handling:

- **Variables** → Data store karte hain (String, Int32, Boolean, Generic, etc.)
- **Scope** → Sequence, Flowchart, ya Global
- **Arguments** → Workflows ke beech data pass karte hain (In, Out, In/Out)
- **Data Handling** → DataTables, Arrays, Lists, Dictionaries structured & dynamic data ke liye

File/Data Operations:

- Most RPA automations files aur data ke saath kaam karti hain.
- Operations: read/write files, filter/sort data, move/copy/archive files.
- Real-world use: invoices, payroll, reports, logs.

Suggested Practice (agar practical question aaye to yeh types practice karo):

- Sequence → Text file read karke date append karna
- Flowchart → Student grading system (Pass/Fail)
- Control Flow → Excel loop karke salary > 40k filter karna
- File Handling → Downloads se Archive mein files move karna timestamp ke saath

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Quick Revision Cheat-Sheet (Last 5 Minute Revision)

- **Workflow structuring benefits** → C-R-E-E-S (Clarity, Reusability, Easier debug, Error handling, Scalability)
- **Sequence vs Flowchart vs State Machine** → Linear vs Branching vs States
- **Control Flow activities** → If (2 options), Switch (3+ options), While/Do-While/For Each (loops), Parallel (simultaneous)
- **Loop types** → While(pre-check), Do While(post-check), For Each(collection), Parallel(independent tasks)
- **Variable types** → String, Int32, Boolean, Double, DateTime, Array, DataTable, Dictionary

- **Variable scope** → Sequence < Flowchart < Global
- **Arguments** → In (andar), Out (bahar), In/Out (dono)
- **Collections** → Array(fixed), List(dynamic), Dictionary(key-value)
- **File operations** → Read (load), Write (overwrite), Append (add at end)
- **Excel/CSV** → Read Range/CSV → DataTable → Filter/Sort/Loop → Write Range/CSV
- **Break vs Continue** → Break(loop exit) vs Continue/Pass(skip iteration)

All the best for exam! 💪 Concepts clear ho jayenge iss revision se.